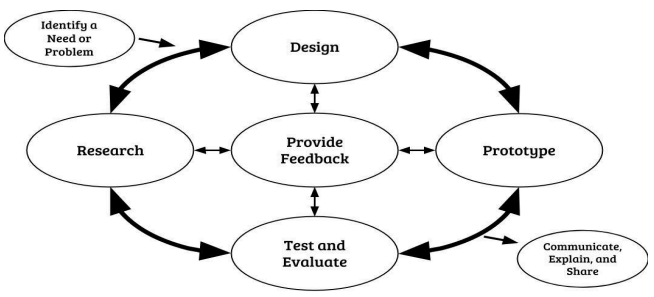




Name _____ HR _____

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6 Science Unit 1 Engineering STAR Cards!

QUESTION	ANSWER
1a)  What is the engineering design process? 	1a) The engineering design process is the set of steps engineers follow when creating solutions to solve a problem
1b)  When developing possible solutions to a problem, engineers need to consider the criteria and constraints of their design solution. What are <u>criteria</u> and <u>constraints</u>?	1b) <u>Criteria</u> = The <u>requirements</u> that make it a good solution <u>Constraints</u> = The <u>limitations</u> you are allowed to work with

QUESTION	ANSWER
1c)  Describe the following steps of the Engineering Design Process: ● Research ● Design ● Prototype	1c) ● <u>Research</u> - gather information to develop ideas. ● <u>Design</u> - take the ideas from research and make a drawing. ● <u>Prototype</u> - take the drawing and make a physical model
1d)  When engineers are deciding which materials to use in their prototypes, they must consider the material's <u>properties</u>. List at least 3 examples of <u>properties</u> engineers may consider. (*Try all 5!*)	1d) ● Flexibility ● Transparency (see through) ● Waterproof ● Thermal Conductor or Insulator ● Electrical Conductor or Insulator